LEGAL SETU
A VOICE FIRST MULTILINGUAL LEGAL ASSISTANCE AND ADVOCATE DISCOVERY PLATFORM

A MINI PROJECT SYNOPSIS SUBMITTED
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Synopsis revision

Updated 5 November 2026. The project is named Legal Setu. The synopsis covers the browser prototype, recent legal-data and retrieval work, and the separate multilingual portal developed in the rebuild workspace. It distinguishes implemented components, recorded test results and remaining integration work.

Scope of the submission

The academic deliverables include the application prototype, requirements, source audit, evaluation evidence and an implementation roadmap. The immediate goal is a reviewable pilot. A production legal service requires further integration, professional review and operational validation.

Key Technical Abbreviations and Nomenclature

Term	Meaning in this project
AI	Artificial intelligence used for preliminary issue analysis; not professional legal advice.
LLM	Large language model invoked through a configured Gemini or gateway adapter.
STT and TTS	Speech-to-text and text-to-speech; availability depends on the selected browser or speech engine.
RAG	Retrieval-augmented generation using retrieved passages as evidence for a generated answer.
BM25	Keyword-based ranking used to retrieve legal passages.
E5 and ONNX	Multilingual embedding model family and model runtime format for local vector retrieval.
RRF	Reciprocal-rank fusion, combining keyword and vector result rankings.
BNS	Bharatiya Nyaya Sanhita, 2023; the indexed reference snapshot needs legal currency review.
API	Application programming interface between clients, application functions and external providers.
RBAC	Role-based access control for citizen, advocate and administrator operations.
HMAC	Hash-based message authentication code used for payment signatures and provider verification.
SHA-256	Cryptographic hash used to detect changes to downloaded source files and indexed content.
MiB	Mebibyte, equal to 1,048,576 bytes; the portal case-attachment limit is 3 MiB per file.
SPA	Single-page application; the React/Vite browser prototype.

1 Project Overview

Legal Setu is a voice-oriented legal assistance and advocate discovery project intended to help citizens describe a concern, understand a preliminary issue category and approach an appropriate legal professional. The platform emphasizes plain language, review of AI output and a clear distinction between information and a confirmed consultation.

The browser prototype uses React and Vite with an Express API and MongoDB. It supports typed intake, Hindi or English browser dictation, follow-up conversation, issue classification, spoken output, advocate profiles and consultation requests. Recent team changes add an animated advocate avatar, repair microphone and profile flows, validate future booking dates and improve payment-signature handling. WhatsApp has been removed from the website navigation; a separate messaging demonstration remains in the backend.

A new legal knowledge subsystem collects official judgment PDFs and a BNS reference text, retains source metadata and retrieves passages with keyword, vector or hybrid search. A grounded-answer endpoint checks passage identifiers and supporting quotations. The browser triage endpoint remains separate from this RAG pipeline; their integration is a remaining milestone [1–4].

The separate Next.js portal in rebuild adds Clerk identity, Convex persistence, citizen/advocate/administrator roles, credential review, private cases, reviewed summaries, consented Gemini chat, attachments and slot-based consultation workflows. English, Hindi and Punjabi are its pilot interface languages. These advances belong to a distinct implementation and are not all present in the Vite application [5].

Project boundaries

The system provides preliminary information, not a legal opinion, guaranteed emergency response or case-outcome prediction. A stored availability flag does not prove a real advocate is available. Payment signature verification does not establish escrow, refunds or a completed consultation.

Browser speech may use a browser vendor’s remote service. Interface localization does not prove recognition, pronunciation or legal comprehension in every language. Credential verification, privacy policy, retention, security review and legal review are required before a public service can be claimed.

2 Problem Statement

Citizens can describe an event in everyday language while still being unable to identify the relevant legal domain, organize supporting facts or choose an appropriate advocate. A useful intake system must reduce this initial uncertainty without disguising automated suggestions as authoritative legal advice.

Information and language barriers

Legal terminology and lengthy forms make it difficult to explain a concern under stress. Typed and spoken intake, editable transcripts and plain-language summaries offer alternative ways to begin. Recognition errors must remain visible and correctable before submission.

Evidence and legal currency

An LLM can produce a plausible explanation with an incorrect section or unsupported conclusion. Retrieval must preserve original sources, page or section references and explicit insufficient-evidence outcomes. A complete section inventory alone does not establish that amendments, commencement and current applicability have been checked.

Trust and consultation status

Users need to distinguish a sample advocate from an approved profile, a request from an accepted appointment and a payment order from a successful payment. The prototype and portal currently implement different parts of this workflow. Production claims require verified onboarding, authorization and a tested service process.

Privacy and integration

Case narratives and attached documents can contain sensitive personal information. Browser storage is persistent local storage, not a compliance guarantee. The portal introduces backend access checks and explicit AI-sharing consent, while retention, deletion, provider terms and operational controls still need a defined policy.

Issue	Current response	Remaining need
Intake friction	Voice or typed input; editable transcript	Real device and human-language evaluation
Unsupported answers	Source retrieval and quotation checks	Reviewed applicability and answer quality
Unclear appointments	Requests and portal lifecycle states	Real advocate acceptance and operational validation
Fragmented implementation	Prototype, RAG backend and portal	Agreed architecture and integration plan

3 Objectives of the Project

The updated objectives extend the original bilingual triage prototype into a source-aware, multilingual and reviewable consultation workflow. Completion is assessed through observable behavior rather than unsupported latency or accuracy guarantees.

Objective	Implementation focus	Verification requirement
1 Accessible intake	Typed entry, Hindi/English dictation, transcript editing, avatar and read-aloud	Permission/error handling; editable output; device and language checks
2 Preliminary analysis	Structured domain, urgency, summary, next steps and follow-up context	Input/output validation; visible provider failure; review before reliance
3 Legal retrieval	Official-source ingestion, BM25, multilingual E5 vectors and hybrid ranking	Source hashes, page/section citations, stale-index checks and reviewed queries
4 Private case workspace	Clerk/Convex roles, saved cases, reviewed summaries, consented chat and files	Ownership enforcement, consent boundaries and persistence checks
5 Advocate workflow	Profile review, discovery, future slots, requests, acceptance and cancellation	Authorization, atomic reservation/release and duplicate-request handling
6 Measured readiness	Functional tests, source audits, language review and pilot scenarios	Publish actual results and failures; separate mocks from live services

Acceptance principles

The citizen must be able to review the transcript and summary, save privately where supported and choose whether case text or selected documents are sent to an AI provider. Failure must not be shown as a successful analysis or booking.

The retrieval objective includes a larger lawyer-reviewed Hindi, English and mixed-language dataset. The existing four-query smoke evaluation is useful diagnostic evidence, not a legal-accuracy benchmark. Payment, emergency dispatch and native mobile delivery remain outside the portal MVP.

4 Proposed Solution and Methodology

Development proceeds through separate, explicit application paths. The existing website offers intake and triage; the knowledge backend adds source-aware retrieval; the portal implements authenticated case and consultation workflows. Consolidation is planned rather than assumed.

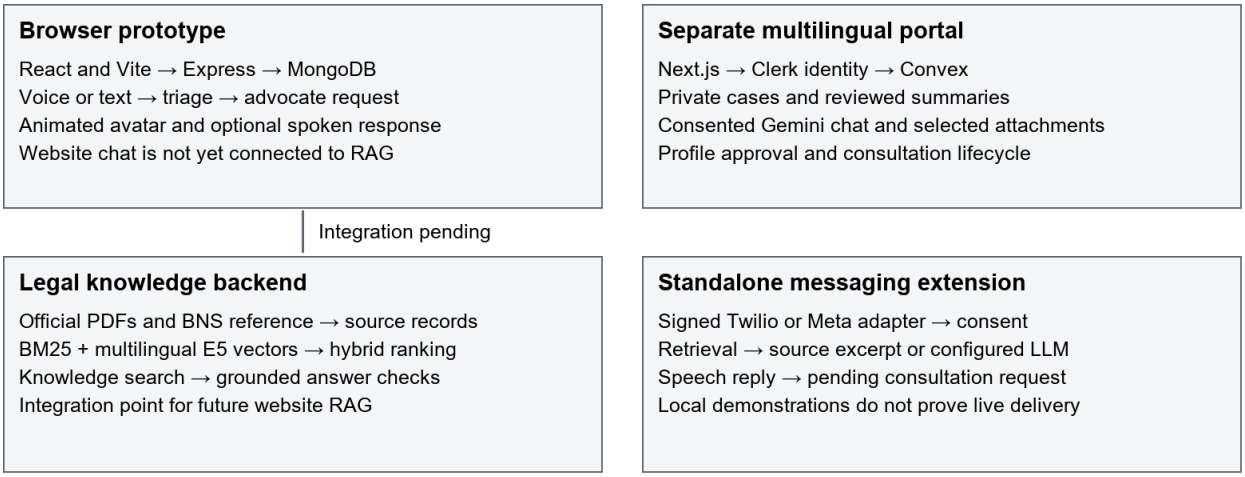


Figure 4.1 Current Legal Setu application boundaries

Processing stages

1. Intake and review. Capture speech or typed text, allow corrections and request only the information needed for the concern. The portal permits private saving without an AI call.

2. Analysis and retrieval. The browser triage API invokes the configured model or rule-based fallback. Separately, knowledge search retrieves passages and validates index freshness; grounded generation checks returned passage IDs and exact source quotations.

3. Private case conversation. In the portal, save reviewed summaries and case history. Send case text and selected files to Gemini only with explicit consent. PDF, JPG, PNG and TXT attachments are limited to 3 MiB each.

4. Consultation lifecycle. Discover eligible advocates and request a future slot. The portal reserves slots atomically and supports acceptance, cancellation, expiry and notifications. The Vite booking record is a simpler request and does not provide the same lifecycle.

5 Key Technologies and Tools

Subsystem	Technology	Scope
Existing website	React 18 and Vite	Browser intake, conversation, avatar, profiles and booking interface
Prototype backend	Node.js, Express, MongoDB and Mongoose	Triage, matching, requests, knowledge endpoints and payment prototype
Speech and avatar	Web Speech API; animated image states and GLB asset	Browser-dependent recognition/playback; visual assistant without guaranteed lip synchronization
Legal ingestion	Fetch, Cheerio, Firecrawl adapter and PDF extraction	Official-source collection, metadata, source hashes, page/section passages
Semantic retrieval	Transformers.js, multilingual-e5-small and ONNX	384-dimensional local CPU embeddings; BM25/vector/RRF hybrid modes
Portal frontend	Next.js, React, TypeScript, Tailwind and shadcn/ui	Separate responsive multilingual application
Portal identity and data	Clerk and Convex	Sessions, backend permissions, private storage and booking state transitions
AI providers	Server-side Gemini; configurable legacy gateway	Preliminary analysis and consented conversation; credentials stay on the server
Development and tests	Bun, Turborepo, native TypeScript compiler and test runners	Portal workspaces, type checks, focused tests and builds
Optional integrations	Razorpay; Twilio/Meta; local Whisper and speech output	Prototype payments and separate messaging experiments

Execution and deployment

The website runs locally on port 5173 with an API on port 5000 and a local MongoDB instance. The separate portal uses its own Next.js server and hosted Clerk/Convex configuration. HTTPS and operational monitoring are deployment requirements, not evidence of a configured production environment.

Embedding inference can run on CPU after model download. Corpus files, vectors and cached models are excluded from Git and must be restored or rebuilt. A fresh clone does not automatically contain the previously audited legal dataset.

6 V E T S Justification

V Viability

The application can be developed and demonstrated on commodity hardware. Browser speech avoids building a custom recognition service for the website, while local embeddings avoid an embedding API dependency at query time. Initial model/database downloads, provider usage, hosting and maintenance still have costs. The separate implementations require an explicit consolidation decision to keep the academic scope manageable.

E Engineering depth

The engineering contribution combines editable speech intake, structured model integration, deterministic fallback, source ingestion and provenance, multilingual embeddings, hybrid retrieval and evidence validation. The portal adds backend-owned permissions, private document access, reviewed summaries, provider consent and atomic consultation-state transitions.

These mechanisms address concrete failure cases: an edited transcript, an unavailable provider, a changed source, a stale vector index, a duplicate request, an unauthorized document read or an expired reservation. Correct behavior in those cases is more meaningful than an unmeasured claim of universal reliability.

T Technology and public service relevance

Legal Setu applies conversational interfaces and source-aware AI to early legal assistance. Its focus on Indian legal materials and English/Hindi/Punjabi portal interfaces is relevant to language accessibility. Legal accuracy, statutory currency and privacy compliance require professional review and defined procedures; they do not follow automatically from the technology stack.

S Social and industry impact

The intended benefit is a clearer first step: citizens can organize a concern, review preliminary information and request professional assistance. This benefit depends on comprehensible language, truthful service status and available advocates. The project does not claim measured reductions in legal costs, court pendency or case duration.

Responsible deployment

Pilot operation should use clearly labelled synthetic data until access controls, source review, retention/deletion, provider terms and real consultation handling have been approved. AI outputs must remain reviewable and must not be presented as judicial conclusions or professional representation.

7 Expected Outcomes and Current Progress

Deliverable	Current progress	Qualification
Browser prototype	Voice/text intake, avatar, triage, profiles and booking code	Local homepage and API health checked; speech and model accuracy need live evaluation
Legal corpus	Recorded audit: 3 judgments, 26 pages, 44 judgment passages; 358 BNS sections, 432 statute passages	476 passages total; corpus is external to Git; current-law certification pending
Hybrid retrieval	BM25, local multilingual embeddings, vector/hybrid modes and evidence checks	Website triage is separate; model/index setup required on a new clone
Multilingual portal	Roles, review, private cases, summaries, chat, documents and consultation lifecycle	Separate implementation; English/Hindi/Punjabi pilot; wider language support incomplete
Messaging extension	Consent, signed adapters, source replies and pending requests	Standalone demonstration; live delivery and real multilingual speech unverified
Payment prototype	Order creation and HMAC verification; improved validation	No verified escrow/refund service; paid bookings deferred in the portal MVP

Recorded evaluation and its limits

The 5 November developer smoke evaluation returned the expected BNS section among the top five results for 3 of 4 queries. The Hindi fraud query missed its expected section. This small developer-labelled sample is not a lawyer-reviewed accuracy score [3].

The source audit records matching hashes for the three official judgment PDFs and reconstruction checks for indexed passages. This verifies source-text integrity; it does not verify legal applicability or every generated conclusion [4].

Further evaluation should cover transcription errors, refusal behavior, multilingual retrieval, source changes, ownership violations, consent withdrawal, concurrent requests and booking expiry. Report functional, live-provider and human-review results separately.

8 Implementation Timeline

The original twelve-week schedule is retained as the academic planning framework. Progress is described by workstream because the prototype, knowledge backend and portal have developed separately. These week allocations are planning estimates, not a reconstructed completion history.

Phase	Workstream	Progress and next gate	Allocation
1	Requirements and scope	Rename to Legal Setu; define application boundaries, roles and evidence requirements. Agree the consolidation path.	Weeks 1–2
2	Accessible browser intake	Existing voice/text intake and avatar; recent voice/profile repairs. Complete real-device speech and accessibility review.	Weeks 3–4
3	Source retrieval and analysis	Ingestion, BM25, vectors and hybrid retrieval implemented. Restore corpus/index; connect website RAG; expand reviewed queries.	Weeks 5–7
4	Private cases and consultations	Portal roles, documents, reviewed summaries and reservation lifecycle implemented. Validate hosted user/advocate/admin workflows.	Weeks 8–10
5	Evaluation and submission	Record actual test outcomes, human language/legal review, privacy controls and final demonstration. Update report and presentation.	Weeks 11–12

Remaining priority milestones

1. Choose the application that will be submitted and integrate the knowledge backend through a reviewed contract. Do not present the separate stacks as one completed deployment.

2. Review corpus currency and add representative Hindi, English and mixed-language legal queries. Measure retrieval and answer quality, including refusal cases and known misses.

3. Validate real identity, private files, consent, advocate approval, slot contention, acceptance and expiry in the selected hosted environment. Define retention, deletion and incident handling.

4. Complete a controlled demonstration using synthetic cases. Treat payment rollout, guaranteed emergency connection, native mobile clients and legal drafting as later work.

9 References

[1] Project repository, i-got-this-faa/Vakil-Setu, main snapshot 30812b9, 5 November 2026. README.md and frontend/backend source. https://github.com/i-got-this-faa/Vakil-Setu

[2] Project documentation, Local multilingual vector retrieval, docs/VECTOR_RETRIEVAL.md, main snapshot 30812b9, 5 November 2026. Describes embedding setup, retrieval modes and integration boundaries.

[3] Project documentation, Retrieval smoke evaluation, docs/RETRIEVAL_EVALUATION.md, generated 5 November 2026. Four developer-labelled queries; expected-section top-five hits in 3 of 4 cases.

[4] Project documentation, RAG data audit, docs/RAG_DATA_AUDIT.md, generated 5 November 2026. Source hashes, passage reconstruction and unresolved BNS currency limitations.

[5] Legal Setu rebuild workspace, README.md and docs/chat-and-documents.md, verification.md, requirements.md and branding-rename.md, inspected 5 November 2026. Source and progress records for the separate multilingual portal.

[6] MDN Web Docs, Web Speech API, accessed 5 November 2026. https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API

[7] L. Wang et al., Multilingual E5 Text Embeddings A Technical Report, 2024; multilingual-e5-small model card. https://huggingface.co/intfloat/multilingual-e5-small

[8] Convex, Developer documentation, accessed 5 November 2026. https://docs.convex.dev/

[9] Razorpay, Node.js Standard Checkout integration steps, accessed 5 November 2026. https://razorpay.com/docs/payments/server-integration/nodejs/integration-steps

[10] Project documentation, docs/IMPLEMENTATION_STATUS.md and docs/TWILIO_SANDBOX.md, main snapshot 30812b9. Historical progress and standalone messaging limits; newer retrieval findings are recorded in references [2–4].

Evidence interpretation

Repository source establishes that an implementation exists. Recorded tests establish only the scenarios and environments exercised. Model documentation describes the model, not Legal Setu’s application accuracy. A proposed requirement or earlier synopsis is not proof of a completed service.